MySQL 8.0 Command Line Cli

```
mysql> -- Q1: Write a query to find the name (first_name, last_name) and the salary of the employees who have a higher salary than the employee whose last_n
ame='Bull'.
mysql> select first_name, last_name, salary from employees where salary > (select salary from employees where last_name = 'Bull');
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q2: Write a query to find the name (first_name, last_name) of all employees who works in the IT department.
mysql> select first_name, last_name from employees where department_id = (select department_id from departments where department_name = 'IT');
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Alexander  | Hunold    |
| Bruce      | Ernst     |
| David      | Austin    |
+------------+-----------+
3 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q3: Write a query to find the name (first_name, last_name) of the employees who have a manager and worked in a USA based department. Hint : Write
single-row and multiple-row subqueries
mysql> select first_name, last_name from employees where manager_id is not null and department_id in (select department_id from departments where location_i
d in (select location_id from locations where country_id = 'US'));
+------------+-------------+
| first_name | last_name   |
+------------+-------------+
| Alexander  | Hunold      |
| Bruce      | Ernst       |
| David      | Austin      |
| Matthew    | Weiss       |
| Adam       | Fripp       |
| Julia      | Nayer       |
| Irene      | Mikkilineni |
| James      | Landry      |
| Steven     | Markle      |
| Jennifer   | Whalen      |
| Neena      | Kochhar     |
| Lex        | De Haan     |
| Nancy      | Greenberg   |
| Daniel     | Faviet      |
| John       | Chen        |
| Ismael     | Sciarra     |
| Jose       | Urman       |
| Shelley    | Higgins     |
| William    | Gietz       |
+------------+-------------+
19 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q4: Write a query to find the name (first_name, last_name) of the employees who are managers.
mysql> select first_name, last_name from employees where employee_id in (select manager_id from employees where manager_id is not null);
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Steven     | King      |
| Neena      | Kochhar   |
| Lex        | De Haan   |
| Alexander  | Hunold    |
| Nancy      | Greenberg |
| Matthew    | Weiss     |
| Adam       | Fripp     |
| John       | Russell   |
| Karen      | Partners  |
| Alberto    | Errazuriz |
| Michael    | Hartstein |
| Shelley    | Higgins   |
+------------+-----------+
12 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q5: Write a query to find the name (first_name, last_name), and salary of the employees whose salary is greater than the average salary.
mysql> select first_name, last_name, salary from employees where salary > (select avg(salary) from employees);
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 24000.00 |
| Neena      | Kochhar   | 17000.00 |
| Lex        | De Haan   | 17000.00 |
| Nancy      | Greenberg | 12008.00 |
| John       | Russell   | 14000.00 |
| Karen      | Partners  | 13500.00 |
| Alberto    | Errazuriz | 12000.00 |
| Peter      | Tucker    | 10000.00 |
| Clara      | Vishney   | 10500.00 |
| Ellen      | Abel      | 11000.00 |
| Michael    | Hartstein | 13000.00 |
| Shelley    | Higgins   | 12008.00 |
+------------+-----------+----------+
12 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q6: Write a query to find the name (first_name, last_name), and salary of the employees whose salary is equal to the minimum salary for their job
grade.
mysql> select first_name, last_name, salary from employees where salary = (select min_salary from jobs where jobs.job_id = employees.job_id);
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q7: Write a query to find the name (first_name, last_name), and salary of the employees who earns more than the average salary and works in any of
 the IT departments.
mysql> select first_name, last_name, salary from employees where salary > (select avg(salary) from employees) and department_id in (select department_id fro
m departments where department_name like 'IT%');
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q8: Write a query to find the name (first_name, last_name), and salary of the employees who earns more than the earning of Mr. Bell.
mysql> select first_name, last_name, salary from employees where salary > (select salary from employees where last_name = 'Bell');
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q9: Write a query to find the name (first_name, last_name), and salary of the employees who earn the same salary as the minimum salary for all dep
artments.
mysql> select first_name, last_name, salary from employees where salary in (select min(salary) from employees group by department_id);
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Neena      | Kochhar   | 17000.00 |
| Lex        | De Haan   | 17000.00 |
| Bruce      | Ernst     |  6000.00 |
| David      | Austin    |  4800.00 |
| Ismael     | Sciarra   |  7700.00 |
| Steven     | Markle    |  2200.00 |
| Sundar     | Ande      |  6400.00 |
| Kimberely  | Grant     |  7000.00 |
| Jennifer   | Whalen    |  4400.00 |
| Pat        | Fay       |  6000.00 |
| William    | Gietz     |  8300.00 |
+------------+-----------+----------+
11 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q10: Write a query to find the name (first_name, last_name), and salary of the employees whose salary is greater than the average salary of all de
partments.
mysql> select first_name, last_name, salary from employees where salary > all (select avg(salary) from employees group by department_id);
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 24000.00 |
+------------+-----------+----------+
1 row in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q11: Write a query to find the name (first_name, last_name) and salary of the employees who earn a salary that is higher than the salary of all th
e Shipping Clerk (JOB_ID = 'SH_CLERK'). Sort the results of the salary of the lowest to highest.
mysql> select first_name, last_name, salary from employees where salary > all (select salary from employees where job_id = 'SH_CLERK') order by salary asc;
+------------+-------------+----------+
| first_name | last_name   | salary   |
+------------+-------------+----------+
| Steven     | Markle      |  2200.00 |
| James      | Landry      |  2400.00 |
| Irene      | Mikkilineni |  2700.00 |
| Julia      | Nayer       |  3200.00 |
| Jennifer   | Whalen      |  4400.00 |
| David      | Austin      |  4800.00 |
| Bruce      | Ernst       |  6000.00 |
| Pat        | Fay         |  6000.00 |
| Sundar     | Ande        |  6400.00 |
| Kimberely  | Grant       |  7000.00 |
| Ismael     | Sciarra     |  7700.00 |
| Jose       | Urman       |  7800.00 |
| Matthew    | Weiss       |  8000.00 |
| John       | Chen        |  8200.00 |
| Adam       | Fripp       |  8200.00 |
| William    | Gietz       |  8300.00 |
| Alexander  | Hunold      |  9000.00 |
| Daniel     | Faviet      |  9000.00 |
| Allan      | McEwen      |  9000.00 |
| Peter      | Tucker      | 10000.00 |
| Clara      | Vishney     | 10500.00 |
| Ellen      | Abel        | 11000.00 |
| Alberto    | Errazuriz   | 12000.00 |
| Nancy      | Greenberg   | 12008.00 |
| Shelley    | Higgins     | 12008.00 |
| Michael    | Hartstein   | 13000.00 |
| Karen      | Partners    | 13500.00 |
| John       | Russell     | 14000.00 |
| Neena      | Kochhar     | 17000.00 |
| Lex        | De Haan     | 17000.00 |
| Steven     | King        | 24000.00 |
+------------+-------------+----------+
31 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q12: Write a query to find the name (first_name, last_name) of the employees who are not supervisors.
mysql> select first_name, last_name from employees where employee_id not in (select manager_id from employees where manager_id is not null);
+------------+-------------+
| first_name | last_name   |
+------------+-------------+
| Bruce      | Ernst       |
| David      | Austin      |
| Daniel     | Faviet      |
| John       | Chen        |
| Ismael     | Sciarra     |
| Jose       | Urman       |
| Julia      | Nayer       |
| Irene      | Mikkilineni |
| James      | Landry      |
| Steven     | Markle      |
| Peter      | Tucker      |
| Allan      | McEwen      |
| Clara      | Vishney     |
| Sundar     | Ande        |
| Ellen      | Abel        |
| Kimberely  | Grant       |
| Jennifer   | Whalen      |
| Pat        | Fay         |
| William    | Gietz       |
+------------+-------------+
19 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q13: Write a query to display the employee ID, first name, last name, and department names of all employees.
mysql> select e.employee_id, e.first_name, e.last_name, d.department_name from employees e left join departments d on e.department_id = d.department_id;
+-------------+------------+-------------+-----------------+
| employee_id | first_name | last_name   | department_name |
+-------------+------------+-------------+-----------------+
|         100 | Steven     | King        | Executive       |
|         101 | Neena      | Kochhar     | Executive       |
|         102 | Lex        | De Haan     | Executive       |
|         103 | Alexander  | Hunold      | IT              |
|         104 | Bruce      | Ernst       | IT              |
|         105 | David      | Austin      | IT              |
|         108 | Nancy      | Greenberg   | Finance         |
|         109 | Daniel     | Faviet      | Finance         |
|         110 | John       | Chen        | Finance         |
|         111 | Ismael     | Sciarra     | Finance         |
|         112 | Jose       | Urman       | Finance         |
|         120 | Matthew    | Weiss       | Shipping        |
|         121 | Adam       | Fripp       | Shipping        |
|         125 | Julia      | Nayer       | Shipping        |
|         126 | Irene      | Mikkilineni | Shipping        |
|         127 | James      | Landry      | Shipping        |
|         128 | Steven     | Markle      | Shipping        |
|         145 | John       | Russell     | Sales           |
|         146 | Karen      | Partners    | Sales           |
|         147 | Alberto    | Errazuriz   | Sales           |
|         150 | Peter      | Tucker      | Sales           |
|         158 | Allan      | McEwen      | Sales           |
|         162 | Clara      | Vishney     | Sales           |
|         166 | Sundar     | Ande        | Sales           |
|         174 | Ellen      | Abel        | Sales           |
|         178 | Kimberely  | Grant       | NULL            |
|         200 | Jennifer   | Whalen      | Administration  |
|         201 | Michael    | Hartstein   | Marketing       |
|         202 | Pat        | Fay         | Marketing       |
|         205 | Shelley    | Higgins     | Accounting      |
|         206 | William    | Gietz       | Accounting      |
+-------------+------------+-------------+-----------------+
31 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q14: Write a query to display the employee ID, first name, last name, salary of all employees whose salary is above average for their departments.

mysql> select employee_id, first_name, last_name, salary from employees e where salary > (select avg(salary) from employees where department_id = e.departme
nt_id);
+-------------+------------+-----------+----------+
| employee_id | first_name | last_name | salary   |
+-------------+------------+-----------+----------+
|         100 | Steven     | King      | 24000.00 |
|         103 | Alexander  | Hunold    |  9000.00 |
|         108 | Nancy      | Greenberg | 12008.00 |
|         109 | Daniel     | Faviet    |  9000.00 |
|         120 | Matthew    | Weiss     |  8000.00 |
|         121 | Adam       | Fripp     |  8200.00 |
|         145 | John       | Russell   | 14000.00 |
|         146 | Karen      | Partners  | 13500.00 |
|         147 | Alberto    | Errazuriz | 12000.00 |
|         174 | Ellen      | Abel      | 11000.00 |
|         201 | Michael    | Hartstein | 13000.00 |
|         205 | Shelley    | Higgins   | 12008.00 |
+-------------+------------+-----------+----------+
12 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id	row_num
101	Neena	Kochhar	NKOCHHAR	555.0101	1997-09-21	AD_VP	17000.00	NULL	100	90	2
103	Alexander	Hunold	AHUNOLD	555.0103	2006-01-03	IT_PROG	9000.00	NULL	102	60	4
105	David	Austin	DAUSTIN	555.0105	2005-06-25	IT_PROG	4800.00	NULL	103	60	6
109	Daniel	Faviet	DFAVIET	555.0109	2002-08-16	FI_ACCOUNT	9000.00	NULL	108	100	8
111	Ismael	Sciarra	ISCIARRA	555.0111	2005-09-30	FI_ACCOUNT	7700.00	NULL	108	100	10
120	Matthew	Weiss	MWEISS	555.0120	2004-07-18	ST_MAN	8000.00	NULL	100	50	12
125	Julia	Nayer	JNAYER	555.0125	2005-07-16	ST_CLERK	3200.00	NULL	120	50	14
127	James	Landry	JLANDRY	555.0127	2007-01-14	ST_CLERK	2400.00	NULL	121	50	16
145	John	Russell	JRUSSELL	555.0145	2004-10-01	SA_MAN	14000.00	0.40	100	80	18
147	Alberto	Errazuriz	AERRAZUR	555.0147	2005-03-10	SA_MAN	12000.00	0.30	100	80	20
158	Allan	McEwen	AMCEWEN	555.0158	2004-08-01	SA_REP	9000.00	0.35	146	80	22
166	Sundar	Ande	SANDE	555.0166	2008-03-24	SA_REP	6400.00	0.10	147	80	24
178	Kimberely	Grant	KGRANT	555.0178	2007-05-24	SA_REP	7000.00	0.15	147	NULL	26
201	Michael	Hartstein	MHARTSTE	555.0201	2004-02-17	MK_MAN	13000.00	NULL	100	20	28
205	Shelley	Higgins	SHIGGINS	555.0205	2002-06-07	AC_MGR	12008.00	NULL	101	110	30

15 rows in set (0.00 sec)

mysql>

MySQL 8.0 Command Line Cli

```
mysql> -- Q16: Write a query to find the 5th maximum salary in the employees table.
mysql> select distinct salary as fifth_max_salary from employees order by salary desc limit 1 offset 4;
+------------------+
| fifth_max_salary |
+------------------+
|         13000.00 |
+------------------+
1 row in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q17: Write a query to find the 4th minimum salary in the employees table.
mysql> select distinct salary as fourth_min_salary from employees order by salary asc limit 1 offset 3;
+-------------------+
| fourth_min_salary |
+-------------------+
|           3200.00 |
+-------------------+
1 row in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q18: Write a query to select last 10 records from a table.
mysql> select * from (select * from employees order by employee_id desc limit 10) t order by employee_id asc;
+-------------+------------+-----------+----------+--------------+------------+------------+----------+----------------+------------+---------------+
| employee_id | first_name | last_name | email    | phone_number | hire_date  | job_id     | salary   | commission_pct | manager_id | department_id |
+-------------+------------+-----------+----------+--------------+------------+------------+----------+----------------+------------+---------------+
|         158 | Allan      | McEwen    | AMCEWEN  | 555.0158     | 2004-08-01 | SA_REP     |  9000.00 |           0.35 |        146 |            80 |
|         162 | Clara      | Vishney   | CVISHNEY | 555.0162     | 2005-11-11 | SA_REP     | 10500.00 |           0.25 |        147 |            80 |
|         166 | Sundar     | Ande      | SANDE    | 555.0166     | 2008-03-24 | SA_REP     |  6400.00 |           0.10 |        147 |            80 |
|         174 | Ellen      | Abel      | EABEL    | 555.0174     | 2004-05-11 | SA_REP     | 11000.00 |           0.30 |        147 |            80 |
|         178 | Kimberely  | Grant     | KGRANT   | 555.0178     | 2007-05-24 | SA_REP     |  7000.00 |           0.15 |        147 |          NULL |
|         200 | Jennifer   | Whalen    | JWHALEN  | 555.0200     | 1995-09-17 | AD_ASST    |  4400.00 |           NULL |        101 |            10 |
|         201 | Michael    | Hartstein | MHARTSTE | 555.0201     | 2004-02-17 | MK_MAN     | 13000.00 |           NULL |        100 |            20 |
|         202 | Pat        | Fay       | PFAY     | 555.0202     | 2005-08-17 | MK_REP     |  6000.00 |           NULL |        201 |            20 |
|         205 | Shelley    | Higgins   | SHIGGINS | 555.0205     | 2002-06-07 | AC_MGR     | 12008.00 |           NULL |        101 |           110 |
|         206 | William    | Gietz     | WGIETZ   | 555.0206     | 2002-06-07 | AC_ACCOUNT |  8300.00 |           NULL |        205 |           110 |
+-------------+------------+-----------+----------+--------------+------------+------------+----------+----------------+------------+---------------+
10 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q19: Write a query to list the department ID and name of all the departments where no employee is working.
mysql> select department_id, department_name from departments where department_id not in (select department_id from employees where department_id is not nul
l);
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q20: Write a query to get 3 maximum salaries.
mysql> select distinct salary from employees order by salary desc limit 3;
+----------+
| salary   |
+----------+
| 24000.00 |
| 17000.00 |
| 14000.00 |
+----------+
3 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q21: Write a query to get 3 minimum salaries.
mysql> select distinct salary from employees order by salary asc limit 3;
+---------+
| salary  |
+---------+
| 2200.00 |
| 2400.00 |
| 2700.00 |
+---------+
3 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q22: Write a query to get nth max salaries of employees.
mysql> select distinct salary from employees e1 where 3 = (select count(distinct salary) from employees e2 where e2.salary >= e1.salary);
+----------+
| salary   |
+----------+
| 14000.00 |
+----------+
1 row in set (0.00 sec)

mysql>
```